

James A. Barron

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Education

Ph.D. Candidate, Astrophysics

Present

Queen's University, Kingston, Canada

Thesis: *Stellar Magnetism Through the Instability Strip*

Supervisor: Gregg A. Wade

Master of Science, Astrophysics

Queen's University, Kingston, Canada

Thesis: *A Systematic Study of the TESS Photometry of the Magnetic O-type Stars*, ([ADS](#))

Supervisor: Gregg A. Wade

Bachelor of Science, Physics (Honours)

Queen's University, Kingston, Canada

Bachelor of Science, Specialization in Mathematics

University of Alberta, Edmonton, Canada

Refereed Publications

1. **Barron, J. A.**, Wade, G. A., Folsom, C. P. (2026), The rotational and magnetic properties of Polaris from long-term spectropolarimetric monitoring, *ApJ*, 1005, 1, 112, ([ADS Link](#))
2. Biswas, A., Folsom, C.P., **Barron, J. A.**, Wade, G.A., Bellotti, S., Triglio, C. (2026), Discovery of a rapidly evolving magnetic field in the M-dwarf YZ Cet and constraints on the magnetic field of its planet YZ Cet b, *ApJ*, 997, 2, 268 ([ADS Link](#))
3. **Barron, J. A.**, Wade, G. A., Holgado, G., & Simón-Díaz, S. (2025), A Magnetic Field Detection in the Massive O-type Bright Giant 63 Oph, *ApJ*, 991, 79, ([ADS Link](#))
4. Wade, G. A., Oksala, M., Neiner, C., Boucher, É., & **Barron, J. A.** (2025), Magnetic Field Monitoring of Four Massive A–F Supergiants, *ApJL*, 988, L38, ([ADS Link](#))
5. Biswas, A., Das, B., **Barron, J. A.**, Wade, G. A., & Holgado, G. (2025), A Nonstop Aurora? The Intriguing Radio Emission from the Rapidly Rotating Magnetic Massive Star HR 5907, *ApJ*, 980, 2, 260, ([ADS Link](#))
6. Frost, A.J., Sana, H., Mahy, L., Wade, G., **Barron, J.**, Le Bouquin, J.-B., Mérand, A., Schneider, F. R. N., Shenar, T., Barbá, R. H., Bowman, D. M., Fabry, M., Farhang, A., Marchant, P., Morrell, N. I., & Smoker, J. V. (2024), A magnetic massive star has experienced a stellar merger, *Science*, 384, 6692, 214-217, ([ADS Link](#))
7. **Barron, J. A.**, Wade, G. A., Evans, N. R., Folsom, C. P., & Neilson, H. R. (2022), Finding magnetic north: an extraordinary magnetic field detection in Polaris and first results of a magnetic survey of classical Cepheids, *MNRAS*, 512, 4021-4030, ([ADS Link](#))

Manuscripts to be Submitted

1. **Barron, J. A.**, Wade, G. A., Folsom, C. P., Kochukhov, O., First magnetic detection in a W Virginis Type II Cepheid: Modelling the shock-affected magnetic signature of κ Pavonis, in preparation for *ApJ*

Conference Proceedings

1. **Barron, J. A.**, Wade, G. A., Holgado, G., & Simón-Díaz, S., Discovery of a Magnetic Field in the O-type Bright Giant 63 Oph, IAUS, 402, In Press.
2. **Barron, J. A.**, Wade, G. A., Folsom, C. P., & Kochukhov, O. (2024), First results of a magnetic survey of classical Cepheids, IAUS, 361, 233-235, ([ADS Link](#))
3. **Barron, J.**, Wade, G. A., Bowman, D. M., David-Uraz, A., Munoz, M. S., Pablo, H., & Simón-Díaz, S. (2020), MOBSTER: Identifying Candidate Magnetic O Stars through Rotational Modulation of TESS Photometry, Proceedings of the Polish Astronomical Society, 11, 226-235, ([ADS Link](#))
4. David-Uraz, A., Neiner, C., Sikora, J., **Barron, J.**, Bowman, D. M., Cerrahoğlu, P., Cohen, D. H., Erba, C., Khalack, V., Kobzar, O., Kochukhov, O., Pablo, H., Petit, V., Shultz, M. E., ud-Doula, A., Wade, G. A. (2020), Magnetic OB[A] Stars with TESS: probing their Evolutionary and Rotational properties - The MOBSTER Collaboration, Proceedings of the conference Stars and their Variability Observed from Space, held in Vienna on August 19-23, 2019, ([ADS Link](#))
5. David-Uraz, A.; Neiner, C.; Sikora, J.; **Barron, J.**; Bowman, D. M.; Cerrahoğlu, P.; Cohen, D. H.; Erba, C.; Kobzar, O.; Kochukhov, O.; Petit, V.; Shultz, M. E.; Ud-Doula, A.; Wade, G. A.; Mobster Collaboration (2020), MOBSTER: Establishing a Picture of Magnetic Massive Stars as a Population, Stellar Magnetism: A Workshop in Honour of the Career and Contributions of John D. Landstreet, held 8-11 July 2019 in London, Canada, ([ADS Link](#))

Community Papers

1. **Barron, J. A.**, Wade, G. A., Bagnulo, S., Biswas, A., Bowman, D., David-Uraz, A., Erba, C., Eze, C. I., Folsom, C. P., Gutteridge, C., Khalack, V., Kritcka, J., Labadie-Bartz, J., Landstreet, J., Natan, T., Neiner, C., Oksala, M., Petit, V., Shultz, M., Marharyta, S., Stanley, P., Xia, Z. (2024), New capabilities for the Gemini Observatory: The case for high resolution optical spectropolarimetry, Gemini Observatory Community Brief Recommendation.

Invited Talks

Instituto de Astrofísica de Canarias , Tenerife, Spain	Oct 2022
<i>Magnetic properties of classical pulsators</i>	

Contributed Talks

MIAPbP Bridging Gaps in High-Mass Star Research Workshop , Garching, Germany	Aug 2026
<i>Magnetism of massive stars: Recent results and open questions</i>	
CASCA 2026: MIL-ion Futures of Astrophysics , Montréal, Québec	June 2026
<i>The rotational and magnetic properties of Polaris: Challenges to standard stellar models</i>	
CFHT Users Meeting , Lac-à-l'Eau-Claire, Canada	May 2025
<i>Stellar magnetism in the instability strip: New magnetic detections in classical and Type II Cepheids</i>	
MiMeS Workshop , Queen's University, Canada	May 2025
<i>Follow-up of MiMeS magnetic O star candidates</i>	
MIAPbP Stellar Magnetic Fields Workshop , Garching, Germany	Oct 2023
<i>Stellar magnetism through the instability strip</i>	

TASC6/KASC13 Workshop, KU Leuven, Belgium

Jul 2022

*A systematic study of the TESS photometry of the magnetic O-type stars***MOBSTER-1 Virtual Conference**

Sep 2021

MOBSTER: Identifying candidate magnetic O Stars through rotational modulation of TESS photometry

Conference Posters

First Author**Barron, J. A.**, Wade, G. A., Holgado, G., & Simón-Díaz, S., Discovery of a Magnetic Field in the O-type Bright Giant 63 Oph, IAU Symposium 402: Massive Stars Across Redshifts in the Era of JWST and Large-Scale Surveys, Ensenada, Mexico, September 2025**Barron, J. A.**, Wade, G. A., Folsom, C. P., & Kochukhov, O., The magnetic fields of pulsating Cepheid variables, Magnetic Fields from Clouds to Stars (Bfields-2024), Tokyo, Japan, March 2024**Barron, J. A.**, Wade, G. A., Folsom, C. P., & Kochukhov, O., First results of a magnetic survey of classical Cepheids, RR Lyrae & Cepheid Stars Conference, La Palma, Spain, September 2022**Barron, J. A.**, Wade, G. A., Folsom, C. P., & Kochukhov, O., First results of a magnetic survey of classical Cepheids, IAU Symposium 361: Massive Stars Near and Far, Ballyconnell, Ireland, May 2022**Barron, J. A. O.**, & Wade, G. A. (2021), TESS photometry of the Of?p stars HD 148937 and LMC 164-2, Poster talk at OBA stars: Variability and Magnetic Fields, Virtual, St. Petersburg, ([ADS Link](#))**Barron, J. A. O.**, Wade, G. A. (2021), Is the magnetic activity of Classical Cepheids modulated by pulsation? The case of eta Aquilae, MOBSTER-1 virtual conference: Stellar variability as a probe of magnetic fields in massive stars, July 2020, ([ADS Link](#))**Co-Author**Biswas, A., Folsom, C. P., **Barron, J. A.**, & Wade, G. A. (2025), Zeeman Doppler Imaging of the Exoplanet Hosting M-dwarf System YZ Cet Reveals an Extreme Short-term Change in the Global Magnetic Field, Canada-France-Hawaii Telescope Users Meeting, Lac-à-l'Eau-Claire, Canada, May, 2025Stacey, Erik; Wade, Gregg A.; **Barron, James**, (2021), Plaskett's Star: variability in the TESS photometric data, MOBSTER-1 Virtual Conference: Stellar variability as a probe of magnetic fields in massive stars, ([ADS Link](#))Wade, G. A., Woodcock, K., Lim, O., Bohlender, D., Monin, D., Sikora, J., **Barron, J.**, Shultz, M., Alecian, E., Neiner, C., & MiMeS Collaboration (2021), Two new magnetic B-type stars from the MiMeS survey, MOBSTER-1 Virtual Conference: Stellar variability as a probe of magnetic fields in massive stars, ([ADS Link](#))

Funding and Awards

Natural Sciences and Engineering Research Council of Canada PGS-D

2023 – 2026

G. Neilson Whyte Graduate Fellowship

2025

Queen's University

Conference Travel Award

2022 – 2025

Queen's University, (3 awards)

Harold M. Cave Graduate Travel Scholarship

2023 – 2025

Queen's University, (3 awards)

R.S McLaughlin Fellowship

2022 – 2023

Queen's University

Carl Reinhardt Fellowship Queen's University	2021 – 2022
Graduate Entrance Tuition Award Queen's University	2019 – 2020
NSERC Undergraduate Research Award Royal Military College of Canada	2019
Dean's Honour List Queen's University	2018 – 2019
Harold M. Cave Undergraduate Travel Scholarship Queen's University	2018

Service and Leadership

Observatory Volunteer , Queen's Observatory	Sep 2023 – Present
Volunteer , Queen's Astronomy on Tap	Sep 2023 – Present
Graduate Student Committee Representative , Canadian Astronomical Society	Sep 2022 – Present
Graduate Oral Presentation and Poster Judge , CASCA 2026	June 2026
Proposal Reviewer , Canadian Time Allocation Committee, CFHT	Apr 2026
Presenter , QUARG Journals Club	Sep 2019 – Mar 2026
Category Judge , Frontenac, Lennox & Addington Science Fair	Mar 2026
Adjudicator , Inquiry@Queen's Undergraduate Research Conference	Mar 2026
Volunteer , Science Literacy Week, McDonald Institute & Queen's Observatory	Oct 2025
Graduate Poster Grader , Canadian Astronomical Society Annual Meeting	Jun 2025
Outreach Volunteer , Science Rendezvous Kingston	May 2025
LOC Member and Session Chair , MiMeS Workshop, Queen's University	May 2025
Outreach Volunteer , Science Rendezvous Kingston	May 2024
Volunteer , Queen's Eclipse Ambassador	Sep 2023 – Apr 2024
Graduate Steering Representative , Queen's Graduate Physics Society	Sep 2022 – Aug 2023
Adjudicator , Inquiry@Queen's Undergraduate Research Conference	Mar 2022
Category Judge , Frontenac, Lennox & Addington Science Fair	Mar 2022
LOC Member and Session Chair , MOBSTER-2 Virtual Conference	Nov 2021
Science Fair Judge , Canada-Wide Science Fair	May 2021
Category Judge , Frontenac, Lennox & Addington Science Fair	Mar 2021
Let's Talk Astrophysics Volunteer , Let's Talk Science, Queen's University	Feb 2021
Outreach Presentation , Greenview Elementary School, Edmonton, Canada	Dec 2020
Conference Volunteer , The Physics of Galaxy Scaling Relations, Queen's University	Jul 2018

Research Mentorship

Undergraduate Research (NSERC USRA) , Royal Military College of Canada <i>Development of software analysis pipeline for Gaia DR4</i>	May 2026 - Present
Undergraduate Research Thesis , Queen's University <i>Inferring the atmospheric velocity gradients of classical Cepheids</i>	Sep 2025 – Apr 2026
Undergraduate Research (NSERC USRA) , Royal Military College of Canada <i>A first BRITE, SMEI and TESS analysis of the suspected merger product θ-Scorpii</i> Presented at the 2024 BRITE Side of Stars Conference, Vienna, Austria, (ADS Link)	May 2024 – Aug 2024
Undergraduate Research (NSERC USRA) , Royal Military College of Canada <i>Searching for slowly rotating magnetic massive stars with Gaia</i>	May 2023 – Aug 2023
Undergraduate Research Project , Royal Military College of Canada <i>Analysis of time-series photometry of a B-type supergiant</i>	May 2021 – Aug 2021

Teaching Experience (Queen's University)

Teaching Assistant Introduction to Physics II (PHYS 116) , 1 semester	Jan 2026 – Apr 2026
Teaching Assistant Experimentation and Design (ASPC 102) , 1 semester	Sep 2025 – Dec 2025
Teaching Assistant The Solar System (ASTR 101) , 4 semesters	Sep 2020 – Apr 2024
Teaching Assistant Introduction to Physics Tutorial (PHYS 115) , 1 semester	Sep 2023 – Dec 2023
Teaching Assistant Stellar Structure and Evolution (PHYS 435) , 2 semesters	Sep 2022 – Dec 2023
Teaching Assistant Physical Processes in Astrophysics (PHYS 315) , 1 semester	Sep 2022 – Dec 2022
Teaching Assistant Stars, Galaxies, and the Universe (ASTR 102) , 3 semesters	Sep 2020 – Dec 2022
Teaching Assistant Introductory Physics Laboratory (PHYS 117) , 3 semesters	Sep 2019 – Apr 2022

Professional Development

Research Project Management Training Canadian Association for Graduate Studies	Jan 2026
Science Leadership and Management Queen's University (semester course)	Dec 2022
Introduction to Astronomical Instrumentation Dunlap Institute Summer School, University of Toronto	Jul 2021
Summer School on High Performance and Technical Computing Centre for Advanced Computing, Queen's University	Aug 2019
CRAQ Summer School on Stellar Astrophysics Centre for Research in Astrophysics of Quebec, McGill University	Jun 2019

Observing Programs

Principal Investigator <i>Finding Magnetic North: A first magnetic map of the classical Cepheid Polaris</i> ESPaDOnS/Canada-France-Hawaii Telescope, 25 hours (6 semesters)	2022 – 2025
<i>Characterization of new FS CMa candidates with GHOST: An overlooked channel for post-merger objects</i> GHOST/Gemini South, 2 hours (GS-2025A-FT-219)	2025

<i>Concluding the first systematic magnetic study of classical Cepheids: probing the low and high mass regions of the instability strip</i>	2024
ESPaDOnS/Canada-France-Hawaii Telescope, 13.5 hours (24BC27)	
<i>A first systematic study of the magnetic fields of classical Cepheid stars</i>	2022 – 2023
HARPS/ESO 3.6-metre Telescope, 5 nights (108.22E7.001, 111.24ZW.001)	
Three nights of observing experience as visiting astronomer at La Silla observatory.	
<i>A first systematic study of the magnetic fields of classical Cepheid stars</i>	2020 – 2022
ESPaDOnS/Canada-France-Hawaii Telescope, 53.6 hours (4 semesters)	
<i>Follow-up observations of two magnetic O star candidates identified in the MiMeS survey</i>	2022
ESPaDOnS/Canada-France-Hawaii Telescope, 7.2 hours (22AC31)	
<i>Confirmation of Magnetic O Star Candidates Identified from Rotational Modulation of TESS Photometry</i>	2020
ESPaDOnS/Canada-France-Hawaii Telescope, 9.6 hours (20BC31)	
Co-Investigator	
<i>Spectropolarimetric Followup of Indirect Magnetospheric Signatures from Multi Wavelength Observation of a Rapidly Rotating Massive Star</i>	2025
ESPaDOnS/Canada-France-Hawaii Telescope, 2 hours (25BC23)	
<i>Monitoring the magnetic fields of the A-type supergiants 13 Mon and eta Leo</i>	2025
ESPaDOnS/Canada-France-Hawaii Telescope, 5.4 hours (25BF11)	
<i>Detailed Study of the Magnetic O-star Binary HD 148937: The First Ever Magnetic O-star System Detected at Sub-GHz Frequencies</i>	2025
Giant Microwave Radio Telescope, 35 hours (49_067)	
<i>Follow-up uGMRT observation of the most rapidly-rotating, non-degenerate, magnetic massive star: HD 142184</i>	2024
Giant Microwave Radio Telescope, 14 hours (46_065)	
<i>Constraining the magnetic field of YZ Cet b through star-planet interactions</i>	2024
SPIRou/Canada-France-Hawaii Telescope, 2.6 hours (24BC28)	
<i>The unusual case of the double star HD 155273: a potential hidden magnetic B-type binary</i>	2022
ESPaDOnS/Canada-France-Hawaii Telescope, 4.6 hours (22AC33)	
<i>Magnetism in the upper Hertzsprung-Russell Diagram: using TESS to characterize the distinct sub-population of magnetic OBA stars and hunt for new magnetic candidates</i>	2022
NASA Astrophysics Data Analysis Program (22-DA_ADAP22-0070)	